Chapter 14 Review

Campbell Biology, AP Edition, 12e

Chapter 14 Review

A purple flower.

Summary of Key Concepts

Advanced placement edition

Why was Mendel's choice of pea plants as an experimental organism so fortuitous in developing a model of inheritance? **(BIG IDEA 3)**

To review key terms, go to the **Vocabulary Self-Quiz**.

Concept 14.1: Mendel used the scientific approach to identify two laws of inheritance

- Gregor Mendel formulated a theory of inheritance based on experiments with garden peas, proposing that parents pass on to their offspring discrete genes that retain their identity through generations. This theory includes two "laws."
- The ****law of segregation** states that genes have alternative forms, or ****alleles.** In a diploid organism, the two alleles of a gene segregate (separate) during meiosis and gamete formation; each sperm or egg carries only one allele of each pair. This law explains the 3:1 ratio of F2 ****phenotypes** observed when ****monohybrids** cross-or self-pollinate. Each organism inherits one allele for each gene from each parent. In ****heterozygotes**, the two alleles are different: Expression of the ****dominant allele** masks the phenotypic effect of the ****recessive allele**. ****Homozygotes** have identical alleles of a given gene and are therefore ****true-breeding**.
- The ****law of independent assortment** states that the pair of alleles for a given gene segregates into gametes independently of the pair of alleles for any other gene. In a cross between ****dihybrids** (individuals heterozygous for two genes), the offspring have four phenotypes in a 9:3:3:1 ratio.

When Mendel did crosses of true-breeding purple- and white-flowered pea plants, the white-flowered trait disappeared from the F1 *generation but reappeared in the* F2 *generation. Use genetic terms to explain why that happened.*

Concept 14.2: Probability laws govern Mendelian inheritance

Image of coin toss representing segregation of alleles into sperm from genotype uppercase R lowercase r. One-half sperm are upper case R (heads); one-half sperm are lowercase r (tails).

- The ******multiplication rule** states that the probability of two or more events occurring together (a statement using *and*) is equal to the product of the individual probabilities of the independent single events. The ******addition rule** states that the probability of an event that can occur in two or more independent, mutually exclusive ways (using *or*) is the sum of the individual probabilities.
- The rules of probability can be used to solve complex genetics problems. A dihybrid or other multicharacter cross is equivalent to two or more independent monohybrid crosses occurring simultaneously. In calculating the chances of the various offspring genotypes from such crosses, each character is first considered separately and then the individual probabilities are multiplied.

DRAW IT *Redraw the Punnett square on the right side of* ***Figure 14.8*** *as two smaller monohybrid Punnett squares, one for each gene. Below each square, list the fractions of each phenotype produced. Use the rule of multiplication to compute the overall fraction of each of the possible dihybrid phenotypes. What is the phenotypic ratio?*

Concept 14.3: Inheritance patterns are often more complex than predicted by simple Mendelian genetics

- Extensions of Mendelian genetics for a single gene:

Example images show, 1, purple flower color, 2, Red, pink, white flower color, 3, Red blood cell with both type A and B carbohydrates.

Parts are as follows.

1. Complete dominance of one allele: P P both uppercase = purple flowers. Uppercase P lowercase p = purple flowers.

2. Incomplete dominance of either allele: C superscript R, C superscript R = red flowers. C superscript R, C superscript W= pink flowers. C superscript W, C superscript W = white flowers.

3. Codominance: Red blood cell with both A and B (circles and triangles) carbohydrates on the surface in an I superscript A, I superscript B individual.

- Extensions of Mendelian genetics for two or more genes:

Epistasis: Example: see Figure 14.12. Polygenic inheritance: Example: see Figure 14.13.

- The expression of a genotype can be affected by environmental influences, resulting in a range of phenotypes. Polygenic characters that are also influenced by the environment are called ******multifactorial** characters.
- An organism's overall phenotype reflects its overall genotype and unique environmental history. Even in more complex inheritance patterns, Mendel's fundamental laws still apply.

Which genetic relationships listed in the first column of the two tables above are demonstrated by the inheritance pattern of the ABO blood group alleles? For each genetic relationship, explain why this inheritance pattern is or is not an example.

Concept 14.4: Many human traits follow Mendelian patterns of inheritance

- Analysis of family ******pedigrees** can be used to deduce the possible genotypes of individuals and make predictions about future offspring. Such predictions are statistical probabilities rather than certainties.

Pedigree showing three generations of a family with respect to widow's peak. Images of two sisters shown in the third generation; one with widow's peak and one without.

Generations are as follows.

First generation (grandparents): Pedigree drawn with symbols as follows from left to right: First grandparents: Shaded Square (uppercase W lowercase w) with line drawn to circle (w w, both lowercase). Space and then second grandparents: square (w w both lowercase) with line drawn to shaded circle (uppercase W lower case w).

Second generation (parents, uncles and aunts): Children of first grandparents: Shaded circle (uppercase W lowercase w), square (w w both lowercase), square (w w both lowercase), shaded square (uppercase W lowercase w). This individual has a line drawn to the first child of the second grandparents, a shaded circle (uppercase W lowercase w). Next is a circle (w w both lowercase).

Third generation: children of the couple in the second generation: Shaded circle (W W both uppercase or uppercase W lowercase w). This individual is shown next to a photograph of a woman with Widow's peak. Next is a circle (w w both lowercase). This individual is shown next to a photograph of a woman with no widow's peak.

Is a widow's peak a dominant or recessive trait? Tips for pedigree analysis: Notice in the third generation that the second-born daughter lacks a widow's peak, although both of her parents had the trait. Such a

pattern indicates that the trait is due to a dominant allele. If it were due to a recessive allele, and both parents had the recessive phenotype (straight hairline), all of their offspring would also have the recessive phenotype.

- Many genetic disorders are inherited as simple recessive traits. Most affected (homozygous recessive) individuals are children of phenotypically normal, heterozygous ****__carriers__.
- The sickle-cell allele has probably persisted for evolutionary reasons: Homozygotes have **sickle-cell disease** ****, but heterozygotes have an advantage because one copy of the sickle-cell allele reduces both the frequency and severity of malaria attacks.

Homozygote with sickle-cell disease. Cascade of events in a homozygous individual that leads to sickled red blood cells.

Homozygote with sickle-cell disease: Weakness, anemia, pain and fever, organ damage. Pair of homologous chromosomes, each having a sickle-cell allele, flows to both sickle-cell hemoglobin proteins, flows with low O_2 to part of a fiber of sickle-cell hemoglobin proteins. Long fibers cause red blood cells to be sickle-shaped and causes sickle-cell disease.

- Lethal dominant alleles are eliminated from the population if affected people die before reproducing. Nonlethal dominant alleles and lethal ones that strike relatively late in life can be inherited in a Mendelian way.
- Many human diseases are multifactorial—that is, they have both genetic and environmental components and do not follow simple Mendelian inheritance patterns.
- Using family histories, genetic counselors help couples determine the probability that their children will have genetic disorders. Genetic testing of prospective parents can reveal whether they are carriers of recessive alleles associated with specific disorders. Ultrasounds and blood tests can screen for certain disorders in a fetus. ****__Amniocentesis__ and ****__chorionic villus sampling__ can diagnose whether a suspected genetic disorder is present in a fetus. Other genetic tests can be performed after birth.

Both members of a couple know that they are carriers of the cystic fibrosis allele. None of their three children have cystic fibrosis, but any one of them might be a carrier. The couple would like to have a fourth child but are worried that he or she would very likely have the disease, since the first three do not. What would you tell the couple? Would it remove some uncertainty from their prediction if they could find out whether the three children are carriers?

*For suggested answers, see*Appendix A.**

Tips for Genetics Problems

- Write down symbols for the alleles. (These may be given in the problem.) When represented by single letters, the dominant allele is uppercase and the recessive is lowercase.
- Write down the possible genotypes, as determined by the phenotype.
- If the phenotype is that of the dominant trait (for example, purple flowers), then the genotype is either homozygous dominant or heterozygous (PP or Pp in this example).
- If the phenotype is that of the recessive trait, the genotype must be homozygous recessive (for example, pp).
- If the problem says "true-breeding," the genotype is homozygous.
- Determine what the problem is asking for. If asked to do a cross, write it out in the form [genotype]×[genotype], using the alleles you've decided on.
- To figure out the outcome of a cross, set up a Punnett square.
- Put the gametes of one parent at the top and those of the other on the left. To determine the allele(s) in each gamete for a given genotype, set up a systematic way to list all the possibilities. (Remember, each gamete has one allele of each gene.) Note that there are 2n possible types of gametes, where *n* is the number of gene loci that are heterozygous. For example, an individual with genotype AaBbCc would produce 23=8 types of gametes. Write the genotypes of the gametes in circles above the columns and to the left of the rows.
- Fill in the Punnett square as if each possible sperm were fertilizing each possible egg, making all of the possible offspring. In a cross of AaBbCc×AaBbCc, for example, the Punnett square would have 8 columns and 8 rows, so there are 64 different offspring; you would know the genotype of each and thus the phenotype. Count genotypes and phenotypes to obtain the genotypic and phenotypic ratios. Because the Punnett square is so large, this method is not the most efficient. See tip 5.
- You can use the rules of probability if a Punnett square would be too big. (For example, see the question at the end of the summary for Concept 14.2 and question 7 below.) You can consider each gene separately (see the section Solving Complex Genetics Problems with the Rules of Probability in Concept 14.2).
- If the problem gives you the phenotypic ratios of offspring but not the genotypes of the parents in a given cross, the phenotypes can help you deduce the parents' unknown genotypes.
- For example, if 12 of the offspring have the recessive phenotype and 12 the dominant, you know that the cross was between a heterozygote and a homozygous recessive.
- If the ratio is 3:1, the cross was between two heterozygotes.

- If two genes are involved and you see a 9:3:3:1 ratio in the offspring, you know that each parent is heterozygous for both genes. Caution: Don't assume that the reported numbers will exactly equal the predicted ratios. For example, if there are 13 offspring with the dominant trait and 11 with the recessive, assume that the ratio is one dominant to one recessive.
- For pedigree problems, use the tips in Figure 14.15 and below to determine what kind of trait is involved.
- If parents without the trait have offspring with the trait, the trait must be recessive and the parents both carriers.
- If the trait is seen in every generation, it is most likely dominant (see the next possibility, though).
- If both parents have the trait, then in order for it to be recessive, all offspring must show the trait.
- To determine the likely genotype of a certain individual in a pedigree, first label the genotypes of all the family members you can. Even if some of the genotypes are incomplete, label what you do know. For example, if an individual has the dominant phenotype, the genotype must be AA or Aa; you can write this as A–. Try different possibilities to see which fits the results. Use the rules of probability to calculate the probability of each possible genotype being the correct one.

Test Your Understanding

For multiple-choice questions, go to the **Practice Test**.

Levels 1-2: Remembering/Understanding

- **DRAW IT** A pea plant heterozygous for inflated pods (*Ii*) is crossed with a plant homozygous for constricted pods (*ii*). Draw a Punnett square for this cross to predict genotypic and phenotypic ratios. Assume that pollen comes from the *ii* plant.
- A man with type A blood marries a woman with type B blood. Their child has type O blood. What are the genotypes of these three individuals? What genotypes, and in what frequencies, would you expect in future offspring from this marriage?
- A man who has six fingers on each hand and six toes on each foot and a woman who has the typical number of digits have a daughter who also has the typical number of digits. Remember that extra digits is a dominant trait. What fraction of this couple's children would be expected to have extra digits?
- **DRAW IT** Two pea plants heterozygous for the characters of pod color and pod shape are crossed. (See Table 14.1.) Draw a Punnett square to determine the phenotypic ratios of the offspring.

Levels 3-4: Applying/Analyzing

- Flower position, stem length, and seed shape are three characters that Mendel studied. Each is controlled by an independently assorting gene and has dominant and recessive expression as indicated in Table 14.1. If a plant that is heterozygous for all three characters is allowed to self-fertilize, what proportion of the offspring would you expect to be each of the following? (Note: Use the rules of probability instead of a huge Punnett square.)

- homozygous for the three dominant traits

- homozygous for the three recessive traits

- heterozygous for all three characters

- homozygous for axial and tall, heterozygous for seed shape

- Hemochromatosis is an inherited disease caused by a recessive allele. If both members of a couple are carriers and they have three children, what is the probability of each of the following?

- All three children are of normal phenotype.

- One or more of the three children have the disease.

- All three children have the disease.

- At least one child is phenotypically normal.

(*Note*: It will help to remember that the probabilities of all possible outcomes always add up to 1.)

- The genotype of F1 individuals in a tetrahybrid cross is ***AaBbCcDd***. Assuming independent assortment of these four genes, what are the probabilities that F2 offspring will have the following genotypes?

- **aabbccdd**

- **AaBbCcDd**

- **AABBCCDD**

- **AaBBccDd**

- **AaBBCCdd**

- What is the probability that each of the following pairs of parents will produce the indicated offspring? (Assume independent assortment of all gene pairs.)
- AABBCC×aabbcc→AaBbCc
- AABbCc×AaBbCc→AAbbCC
- AaBbCc×AaBbCc→AaBbCc
- aaBbCC×AABbcc→AaBbCc
- Keisha and Jerome each have a sibling with sickle-cell disease. Neither Keisha nor Jerome nor any of their parents have the disease, and none of them have ever been tested to see if they carry the sickle-cell allele. Based on this incomplete information, calculate the probability that if Keisha and Jerome have a child, the child will have sickle-cell disease.
- In 1981, a stray black cat with unusual rounded, curled-back ears was adopted by a family in California. Thousands of descendants of the cat have since been born, and cat fanciers have developed the curl cat into a show breed. Suppose you owned the first curl cat and wanted to develop a true-breeding variety. How would you determine whether the curl allele is dominant or recessive? How would you obtain true-breeding curl cats? How could you be sure they are true-breeding?

A cat with ears that curl back.

- In tigers, a recessive allele of a particular gene causes both an absence of fur pigmentation (a white tiger) and a cross-eyed condition. If two phenotypically normal tigers that are heterozygous at this locus are mated, what percentage of their offspring will be cross-eyed? What percentage of tigers with crossed eyes will also be white?
- In maize (corn) plants, a dominant allele I inhibits kernel color, while the recessive allele i permits color when homozygous. At a different locus, the dominant allele P causes purple kernel color, while the homozygous recessive genotype pp causes red kernels. If plants heterozygous at both loci are crossed, what will be the phenotypic ratio of the offspring?
- The pedigree below traces the inheritance of alkaptonuria, a biochemical disorder. Affected individuals, indicated here by the colored circles and squares, are unable to metabolize a substance called alkapton, which colors the urine and stains body tissues. Does alkaptonuria appear to be caused by a dominant allele or by a recessive allele? Fill in the genotypes of the individuals whose genotypes can be deduced. What genotypes are possible for each of the other individuals?

Pedigree showing four generations of a family with respect to alkaptonuria.

Squares represent males; circles females. Shaded circles or squares are affected individuals. Lines between circles and squares represent mating. Oldest child of a couple is farthest left. Symbols described from left to right:

First generation: Square (Jorge) with line drawn to shaded circle (Ariana).

Second generation: Children of Jorge and Ariana and their spouses.

Circle (Mariposa) with a line drawn to oldest child of Jorge and Ariana: Shaded square (Benito). Then square (Diego), shaded circle (Elena), (a circle) Carmen with line drawn to square (Miguel).

Third generation: Children of Mariposa and Benito:

Square (Hector) then Square (Julio) with line drawn to circle (Paloma). Children of Carmen and Miguel: Shaded circle (Carlota).

Fourth generation: Children of Julio and Paloma: Square (Roberto).

- Imagine that you are a genetic counselor, and a couple planning to start a family comes to you for information. From a previous marriage, Charles has a child with cystic fibrosis. Charles is currently married to Elaine, whose brother died of cystic fibrosis. What is the probability that Charles and Elaine will have a baby with cystic fibrosis? (Neither Charles, Elaine, nor their parents have cystic fibrosis.)

Levels 5-6: Evaluating/Creating

AP Assessment Questions

- **CONNECT TO BIG IDEA 1** Over the past half century, there has been a trend in the United States and other developed countries for people to marry and start families later in life than did their parents and grandparents. **Describe** what effects this trend might have on the incidence (frequency) of late-acting dominant lethal alleles in the population.

- **SCIENTIFIC INQUIRY** ***Science Practices 3 & 6*** You are handed a mystery pea plant with tall stems and axial flowers and asked to determine its genotype as quickly as possible. You know that the allele for tall stems (T) is dominant to that for dwarf stems (t) and that the allele for axial flowers (A) is dominant to that for terminal flowers (a).

- **Identify** all the possible genotypes for your mystery plant.

- **Describe** the one cross you would do in your garden to determine the exact genotype of your mystery plant.

- While waiting for the results of your cross, you predict the results for each possible genotype listed in part a. **Explain** how you do this and why this is not called “performing a cross.”
- **Explain** how the results of your cross and your predictions will help you learn the genotype of your mystery plant.
- **CONNECT TO BIG IDEA 3** The continuity of life is based on heritable information in the form of DNA. In a short essay (100–150 words), **explain** how the passage of genes from parents to offspring, in the form of particular alleles, ensures perpetuation of parental traits in offspring and, at the same time, genetic variation among offspring. Use genetic terms in your explanation.
- **SYNTHESIZE YOUR KNOWLEDGE*Science Practice 6***

Image of a man wearing a shirt with horizontal stripes, a woman wearing a shirt with vertical stripes, and a child wearing a shirt with both horizontal and vertical stripes.

- Just for fun, imagine that “shirt-striping” is a phenotypic character caused by a single gene. **Construct** a genetic explanation for the appearance of the family in the above photograph, consistent with their “shirt phenotypes.” Include in your answer the presumed allele combinations for “shirt-striping” in each family member. **Identify** the inheritance pattern shown by the child.

*For selected answers, see*Appendix A.**